



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Class : 12

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UNIT TEST -1, JULY - 2024

Time Allowed : 1.30 Hours

MATHEMATICS

[Max. Marks : 50]

Part - I

1. Answer all the questions by choosing the correct answer from the given 4 alternatives.
2. Write question number, correct option and corresponding answer. 10x1=10
3. Each question carries 1 mark.

1. If A is a 3x3 non-singular matrix such that $AA^T = A^T A$ and $B = A^{-1}A^T$, then $BB^T =$
 a) A b) B c) I_3 d) B^T
2. If A, B and C are invertible matrices of some order, then which one of the following is not true?
 a) $\text{adj } A = |A| A^{-1}$ c) $\text{adj}(AB) = (\text{adj } A)(\text{adj } B)$
 b) $A^{-1} = (\det A)^{-1}$ d) $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$
3. If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$, be such that $\lambda A^{-1} = A$, then λ is
 a) 17 b) 14 c) 19 d) 21
4. If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, then $\text{adj}(\text{adj } A)$ is
 a) $\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$ b) $\begin{bmatrix} 6 & -6 & 8 \\ 4 & -6 & 8 \\ 0 & -2 & 2 \end{bmatrix}$ c) $\begin{bmatrix} -3 & 3 & -4 \\ -2 & 3 & -4 \\ 0 & 1 & -1 \end{bmatrix}$ d) $\begin{bmatrix} 3 & -3 & 4 \\ 0 & -1 & 1 \\ 2 & -3 & 4 \end{bmatrix}$
5. The value of $\sum_{i=1}^{13} (i^n + i^{n-1})$ is
 a) $-1+i$ b) i c) 1 d) 0
6. If $|z| = 1$, then the value of $\frac{1+z}{1+\bar{z}}$ is
 a) z b) $\bar{z}$ c) $1/z$ d) 1
7. The principal argument of $\frac{3}{-1+i}$ is
 a) $-\frac{5\pi}{6}$ b) $-\frac{2\pi}{3}$ c) $-\frac{3\pi}{4}$ d) $-\frac{\pi}{2}$
8. If f and g are polynomials of degrees m and n respectively, and if $h(x) = (f \circ g)(x)$, then the degree of h is
 a) mn b) m + n c) m^n d) n^m
9. A polynomial equation in x of degree n always has
 a) n distinct roots b) n real roots c) n imaginary roots d) at most one root.
10. Product of all the roots of n^{th} roots of unity is
 a) 1 b) 0 c) $(-1)^{n-1}$ d) $(-1)^n$

PART - II

1. Answer any 4 questions. 2. Each question carries 2 marks
3. Question number 16 is compulsory

4x2=8

11. Find a matrix A if $\text{adj}(A) = \begin{bmatrix} 7 & 7 & -7 \\ -1 & 11 & 7 \\ 11 & 5 & 7 \end{bmatrix}$



12. Solve the following system of linear equations by matrix inversion method: $2x - y = 8$, $3x + 2y = -2$
13. If $\frac{z+3}{z-5i} = \frac{1+4i}{2}$, find the complex number z in the rectangular form
14. Which one of the points $10 - 8i$, $11 + 6i$ is closest to $1 + i$.
15. Solve the equation $3x^3 - 16x^2 + 23x - 6 = 0$ if the product of the two roots is 1.
16. If A and B are non-singular matrices of order n , then prove that $\text{adj}(AB) = (\text{adj } B)(\text{adj } A)$

PART - III

1. Answer any 4 questions. 2. Each question carries 3 marks
3. Question number 22 is compulsory

4x3=12

17. Find the rank of the matrix by row reduction method:
- $$\begin{bmatrix} 3 & -8 & 5 & 2 \\ 2 & -5 & 1 & 4 \\ -1 & 2 & 3 & -2 \end{bmatrix}$$
18. A chemist has one solution which is 50% acid and another solution which is 25% acid. How much each should be mixed to make 10 litres of a 40% acid solution? (Use Cramer's rule to solve the problem).
19. Find the value of the real numbers x and y , if the complex number $(2+i)x + (1-i)y + 2i - 3$ and $x + (-1+2i)y + 1 + i$ are equal.
20. If the equations $x^2 + px + q = 0$ and $x^2 + p'x + q' = 0$ have a common root, show that it must be equal to $\frac{pq'-p'q}{q-q'}$ or $\frac{q-q'}{p'-p}$
21. Find the square root of $6-8i$.
22. For $n \in \mathbb{N}$, $(1 + \cos \theta + i \sin \theta)^n + (1 + \cos \theta - i \sin \theta)^n = (2)^{n+1} \cos^n \frac{\theta}{2} \cos \frac{n\theta}{2}$

PART - IV

1. Answer all the questions. 2. Each question carries 5 marks

4x5=20

23. a) Find the inverse of each of the following by Gauss-Jordan method:
- $$\begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 6 & -2 & -3 \end{bmatrix}$$
- (OR)
- b) The upward speed $v(t)$ of a rocket at time t is approximated by $v(t) = at^2 + bt + c$, $0 \leq t \leq 100$ where a , b and c are constants. It has been found that the speed at times $t = 3$, $t = 6$, and $t = 9$ seconds are respectively, 64, 133, and 208 miles per second respectively. Find the speed at time $t = 15$ seconds. (Use Gaussian elimination method.)
24. a) Test for consistency and if possible, solve the following systems of equations by rank method.
- $$3x + y + z = 2, x - 3y + 2z = 1, 7x - y + 4z = 5$$
- (OR)
- b) By using Gaussian elimination method, balance the chemical reaction equation: $C_6H_8 + O_2 \rightarrow CO_2 + H_2O$
25. a) Show that the points 1 , $\frac{-1}{2} + i\frac{\sqrt{3}}{2}$, and $\frac{-1}{2} - i\frac{\sqrt{3}}{2}$ are the vertices of an equilateral triangle.
- (OR)
- b) If $z = x + iy$ is a complex number such that $\text{Im}\left(\frac{2z+1}{iz+1}\right) = 0$, show that the locus of z is $2x^2 + 2y^2 + x - 2y = 0$.
26. a) If $\frac{1+z}{1-z} = \cos 2\theta + i \sin 2\theta$, show that $z = i \tan \theta$.
- (OR)
- b) Find a polynomial equation of minimum degree with rational coefficients having $\sqrt{5} - \sqrt{3}$ as a root.